Le Nguyen Gia Hung

AI undergraduate researcher | Explainable AI & multimodal learning
Research interests: trustworthy AI, time-series forecasting and computer vision.
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Portfolio

Education

FPT University — B.Eng. in Artificial Intelligence 2023 – Dec 2027 (expected)
GPA: **8.62/10.0**. Certificate of Merit: semesters 3, 4, 5 and 6.

Publications & Manuscripts

Peer-reviewed conference publication

Gia-Hung Nguyen Le, Gia-Bao Pham Hoang, Tan-Phat Vo, Thu Le and Nhu Nguyen. "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *AJCAI 2025, Springer LNAI*. First and corresponding author. DOI: 10.1007/978-981-95-4969-6_34.

Manuscripts under review

Gia-Hung Nguyen Le, Thu Le and Huynh Quoc Viet. "Distribution-Free Coverage of Attack Classes on Devices Absent from Training." *AISTATS 2027, under review*. First author.

Gia-Hung Nguyen Le, Quoc-Viet Huynh and Thu Le. "Auditing Multi-Source Cyanobacterial Bloom Forecasts under Sparse Environmental Telemetry." *ACIIDS 2027, under review*. First and corresponding author.

Thu Le, **Le Nguyen Gia Hung**, Huynh Quoc Viet, Nguyen Hoai My and Trieu Vuong Nguyen. "Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution." *ACM SIGKDD 2027, under review*. Second author.

Thu Le, **Gia-Hung Nguyen Le**, Hoang Pham Gia Bao and Huynh Quoc Viet. "Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia." *AJCAI 2026, under review*. Second author.

Selected Research Experience

SmokeNet: Air Quality Forecasting Jun – Nov 2025
First & Corresponding Author; published at AJCAI 2025 Academic research

- Designed a Transformer using 48 hours of history to forecast next-day mean $PM_{2.5}$, with seasonal embeddings and reconstruction loss.
- Reported MAE **0.7470** $\mu g/m^3$, R^2 **0.9545** and **35.9% lower MAE than XGBoost** on the held-out 2024 test year.
- Examined forecasts with SHAP, Integrated Gradients and DiCE; a separate May 2024 smoke-event backtest reported **57.7% lower MAE**. Code.

Cyanobacterial Bloom Forecast Audit Jan 2026 – Present
First & Corresponding Author; ACIIDS 2027 under review Academic research

- Co-authored a past-only audit across **7 New Zealand lakes**, evaluating 7- and 14-day forecasts on **170 observed holdout targets** per horizon.
- Full fusion showed no significant gain over Chl-a history after Holm correction; provenance checks reduced 22 apparent episode starts to 3 eligible incidents.

Device-Held-Out Conformal Prediction
First Author; AISTATS 2027 under review Academic research

- Co-authored a device-disjoint conformal prediction study on CIC IoT-DIAD 2024: **40 training, 14 calibration and 14 test devices**; compared device-weighted and device-quantile rules. Code.

Research experience continued

Trustworthy Multimodal Anomaly Detection Jan 2026 – Present
Second Author; ACM SIGKDD 2027 under review Academic research

- Co-authored a leakage audit across **9 domains** and the C-DLE framework for calibrated false-alarm control; reported **94.3% attribution accuracy** against ground-truth anomalies.

Other research. Second author on health-centred air-quality alerts: the study reported 1-day-ahead MAE 3.19 μg/m^3 and $R^2 = 0.54$ in Canterbury.

MERR-GAT: Multimodal Emotion Recognition Oct 2025 – Present
Second & Corresponding Author Research in progress

- Develop audio–text emotion recognition with **WavLM + Compact Conformer**, **BERT/RoBERTa** and GAT fusion; evaluation on IEMOCAP, RAVDESS and MELD is ongoing.

Vietnamese Speech Recognition Sep 2025 – Present
Co-Researcher & Co-Author Research in progress

- Develop a **5.4M-parameter** CNN–BiLSTM/CTC model on **745 hours** of speech; reported **WER 33.09%** and **CER 15.28%**.

Selected course project: Built 9-class PPE detection with YOLOv8/OpenCV, compliance logic and a Streamlit webcam interface with violation reports.

Experience & Leadership

Outlier Aug – Oct 2026
AI Evaluation Contributor Scale AI platform

- Compared two conversational voice models through live Vietnamese interactions, rating naturalness, engagement, audio quality, helpfulness and conversational dynamics; wrote rationales in English.

First Cloud AI Journey — NutriTrack Jan – May 2026
FCAJ Cloud Intern, Team NeuraX AWS Vietnam program

- Developed recipe-generation and AI-coaching workflows using **Amazon Bedrock and AWS Lambda**, structured JSON responses and DynamoDB user context.
- Collaborated on AppSync integration; documented timeout handling and workshops. Report.

SpeedyLabX Student Research Group Jan 2025 – Present
Founder & Student Lead FPT University

- Lead a **10-member** undergraduate group through technical workshops, paper reading and shared research and submission workflows.

Hackathon Projects & Competitions

BioListen VN — Vietnam AI Innovation Challenge Jul 2026
UI & LLM; placed 6th in Round 1 Team NeuraX.ai

- Built monitoring views for audio input, spectrograms and alerts; integrated API data and Vietnamese ranger-report prompts. Next.js, TypeScript, FastAPI/Groq stack. Code.

CosmicOptic — NASA Space Apps Challenge Oct 2025
Frontend & Backend Developer Hackathon participant

- Built React/D3 light-curve views and a FastAPI backend; demonstrated the inference interface with synthetic signals. Code.

Also participated: FPT Hackathon 2025 (algorithmic programming); FPT University AI Hackathon (TeamFinder, a course-teammate website).

Skills

Languages & frameworks: Python, SQL/PostgreSQL, TypeScript, React, FastAPI.
ML & evaluation: PyTorch, Transformers, TensorFlow/Keras, Scikit-learn, OpenCV; SHAP, Integrated Gradients, DiCE, chronological evaluation.
Data & tools: Pandas, NumPy, Git, Docker, Linux; Bedrock, Lambda, DynamoDB, AppSync.